## Topic 1 – Key concepts in biology

| Students should: | | Maths skills |
|---|---|---|
| 1.1 | Explain how the sub-cellular structures of eukaryotic and prokaryotic cells are related to their functions, including:<br>a animal cells – nucleus, cell membrane, mitochondria and ribosomes<br>b plant cells – nucleus, cell membrane, cell wall, chloroplasts, mitochondria, vacuole and ribosomes<br>c bacteria – chromosomal DNA, plasmid DNA, cell membrane, ribosomes and flagella | |
| 1.2 | Describe how specialised cells are adapted to their function, including:<br>a sperm cells – acrosome, haploid nucleus, mitochondria and tail<br>b egg cells – nutrients in the cytoplasm, haploid nucleus and changes in the cell membrane after fertilisation<br>c ciliated epithelial cells | |
| 1.3 | Explain how changes in microscope technology, including electron microscopy, have enabled us to see cell structures and organelles with more clarity and detail than in the past and increased our understanding of the role of sub-cellular structures | |
| 1.4 | Demonstrate an understanding of number, size and scale, including the use of estimations and explain when they should be used | 1d<br>2h |
| 1.5 | Demonstrate an understanding of the relationship between quantitative units in relation to cells, including:<br>a milli ($10^{-3}$)<br>b micro ($10^{-6}$)<br>c nano ($10^{-9}$)<br>d pico ($10^{-12}$)<br>**e calculations with numbers written in standard form** | 1b<br>2a<br>2h |
| 1.6 | *Core Practical: Investigate biological specimens using microscopes, including magnification calculations and labelled scientific drawings from observations* | 1d<br>2a, 2h<br>3b |
| 1.7 | Explain the mechanism of enzyme action including the active site and enzyme specificity | |
| 1.8 | Explain how enzymes can be denatured due to changes in the shape of the active site | |
| 1.9 | Explain the effects of temperature, substrate concentration and pH on enzyme activity | 2c, 2f<br>4a, 4c |

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|---|---|---|
| 1.10 | *Core Practical: Investigate the effect of pH on enzyme activity* | 2c, 2f<br>4a, 4c |
| 1.11 | Demonstrate an understanding of rate calculations for enzyme activity | 1a, 1c |
| 1.12 | Explain the importance of enzymes as biological catalysts in the synthesis of carbohydrates, proteins and lipids and their breakdown into sugars, amino acids and fatty acids and glycerol | |
| 1.13B | *Core Practical: Investigate the use of chemical reagents to identify starch, reducing sugars, proteins and fats* | |
| 1.14B | Explain how the energy contained in food can be measured using calorimetry | 1a<br>2a |
| 1.15 | Explain how substances are transported into and out of cells, including by diffusion, osmosis and active transport | |
| 1.16 | *Core Practical: Investigate osmosis in potatoes* | 1c<br>2b, 2f<br>4a, 4c |
| 1.17 | Calculate percentage gain and loss of mass in osmosis | 1a, 1c<br>4a, 4c |

**Use of mathematics**

- Demonstrate an understanding of number, size and scale and the quantitative relationship between units (2a and 2h).
- Use estimations and explain when they should be used (1d).
- Carry out rate calculations for chemical reactions (1a and 1c).
- **Calculate with numbers written in standard form (1b).**
- Plot, draw and interpret appropriate graphs (4a, 4b, 4c and 4d).
- Translate information between numerical and graphical forms (4a).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Use a scatter diagram to identify a correlation between two variables (2g).
- Understand and use simple compound measures such as the rate of a reaction (1a and 1c).
- Calculate the percentage gain and loss of mass (1c).
- Use fractions and percentages (1c).
- Calculate arithmetic means (2b).
- Carry out rate calculations (1a and 1c).

**Suggested practicals**

- Investigate the effect of different concentrations of digestive enzymes, using and evaluating models of the alimentary canal.
- Investigate the effect of temperatures and concentration on enzyme activity.
- Investigate plant and animal cells with a light microscope.
- Investigate the effect of concentration on rate of diffusion.